

Plastics can be used more sustainably in agriculture

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Plastics have become an integral component in agricultural production as mulch films, nets, storage bins and in many other applications, but their widespread use has led to the accumulation of large quantities in soils. Rational use and reduction, collection, reuse, and innovative recycling are key measures to curb plastic pollution from agriculture. Plastics that cannot be collected after use must be biodegradable in an environmentally benign manner. Harmful plastic additives must be replaced with safer alternatives to reduce toxicity burdens and included in the ongoing negotiations surrounding the United Nations Plastics Treaty. Although full substitution of plastics is currently not possible without increasing the overall environmental footprint and jeopardizing food security, alternatives with smaller environmental impacts should be used and endorsed within a clear socio-economic framework. Better monitoring and reporting, technical innovation, education and training, and social and economic incentives are imperative to promote more sustainable use of plastics in agriculture.

The global population surpassed 8 billion in November 2022 and is expected to increase to about 10 billion by 2050, further increasing the challenge of securing food for humans.

Over the past 20 years, innovation in agricultural technologies has increased the Earth's overall capacity to provide food for more people. However, agriculture exerts one of the greatest pressures on the environment, and current practices frequently conflict with the United Nations

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Sustainable Development Goals (SDGs)¹. While food production needs to increase further, agriculture is already responsible for 29% of greenhouse gas (GHG) emissions, 30% of energy consumption, 33% of land use, 70% of groundwater extraction, and 75% of deforestation². Global warming, associated with many of these factors, negatively impacts crop yield. Linking agricultural intensification with ecosystem protection and spurring technological innovation is key to raising productivity³. In recent decades, plastics have played an ever-increasing role in achieving these goals to the point that they have become an integral component of modern plant agriculture. However, it is important that the short-term benefits of plastic use do not compromise long-term sustainability.

The aim of this review is to present a comprehensive and balanced assessment of the advantages and drawbacks associated with the utilization of plastics in agriculture, with a specific focus on plant agriculture. In addition to inspecting current applications, benefits, adverse effects, and risks, we specifically address the requirements for technological advancements, incentives and regulations, and social processes that could contribute to mitigating plastic pollution and identify pathways toward more sustainable use of plastics in agricultural practices.

Plastic is a generic term for a material based on one or more organic polymers and containing additional substances (additives) for the desired material properties, for example, flame retardants, antistatics, coloring and spinning agents, stabilizers, reinforcing materials and fillers, UV-adsorbents, and plasticizers. Conventional polymers, such as polyethylene (PE), polypropylene (PP), and polyvinyl chloride (PVC) are environmentally persistent due to the high chemical stability of their polymer backbones. Agricultural plastics composed of these conventional polymers can physically fragment into micro- and nanoplastics (MNP), which can accumulate in soils over time, be taken up by biota, or be transported into adjacent receiving environments. In contrast, biodegradable polymers are designed to undergo microbial metabolic utilization, a process in which the polymer carbon is converted to CO₂ and microbial biomass under oxic conditions. The biodegradability of a polymer depends not only on its physicochemical properties but also on the environmental conditions in which the polymer degrades. Although biodegradable polymers are designed to be ultimately completely mineralized to CO₂, the process may also entail the transient formation of ever-smaller MNP fragments. Biodegradable (and conventional) polymers can be from non-renewable petrochemical or renewable bio-based sources⁴. For example, polybutylene adipate-co-terephthalate is produced from petrochemical sources, while starch, cellulose, chitosan, and poly(lactic acid) are derived from renewable sources. While this review focuses mainly on structural polymers, non-structural polymers (polymers in liquid formulation, water-soluble polymers) are also widely used in agriculture, e.g., for encapsulation or dispersion of pesticides, herbicides and nutrients, seed coatings, and soil conditioners. In general, structural polymers can potentially be collected after the intended use, reused, or recycled, while non-structural polymers cannot, as they dissipate into the environment. However, the key concepts presented here also apply to non-structural polymers.

There are multiple applications of plastics in agriculture and their use has increased over the last 70 years to an estimated 12.5 million tons annually⁵, which has led to the accumulation of large quantities of macro-, micro-, and nanoplastics in soils and other receiving environments. The pervasive use of plastics in agriculture poses a growing risk to soil functions, and the wider natural environment, highlighting the urgency for a more sustainable use of plastics in agricultural food production. In March 2022, 175 countries agreed to negotiate a global and legally binding Plastics Treaty (UNEA-5.2) to end plastic pollution by

2024^{6–8}. This international policy instrument aims to address the ecological and human health risks posed by the entire plastic lifecycle, including those linked to agriculture. It is anticipated that regular reporting will be required by governments and non-government stakeholders on the implementation and performance of their goals to reduce plastic pollution⁹.

Current applications and benefits

Plastics in plant agriculture have many environmental and societal benefits^{5,10} (Fig. 1). Plastic mulch films, which alone account for ~50% of the mass of all agricultural plastics, are widely used in crop production¹¹. They provide multiple agronomic benefits, including weed and pest control, soil moisture conservation, a means to control soil and air temperatures, and enhanced nutrient uptake. All these benefits translate to an increase in yield, improved water and nutrient use efficiency, and reduced pesticide use. In China, for example, without the use of mulch film, an additional 3.9 million hectares of arable land would be required to produce the same amount of food¹². The increased soil temperature below plastic mulch films allows farmers to plant and harvest crops earlier and thus provides a market benefit. Plastic mulch films are also used in organic agriculture because they help suppress weed and insect infestation without the need to apply synthetic pesticides.

Conventional plastic mulch films are typically composed of low-density PE, but can also be made of other polymers such as PVC or ethylene-vinyl acetate copolymers. At 40–50 kg/ha, the use of plastic mulch films is highest in Europe and Asia, while it is slightly lower in North and South America at 10–20 kg/ha⁵. Incomplete collection after use leads to the accumulation of persistent plastic residues in soils. Furthermore, chemical additives can leach out from the mulch films. Thus, over repeated mulch applications, the accumulation of plastic residues and released additives can lead to detrimental effects on soil productivity and soil health. Certified soil-biodegradable mulch films are marketed as alternatives to thin (<20–25 µm) conventional mulch films. These biodegradable films can be ploughed into the soils after harvest, where under oxic conditions they are intended to completely biodegrade into CO₂ and microbial biomass¹³.

Plastic sheets are standard covering materials in greenhouses¹⁴ and high and low tunnels, providing thermal insulation, radiant energy capture, and protection against weather and pests. They are produced from a range of polymers and contain additives that provide diverse optical and energetic properties. Plastics are also used as shade and protective nets¹⁵ (e.g., sunshades, anti-hail, anti-bird covers) or as seedling plugs.

Food security depends on nitrogen fertilizers, but their production and use account for 5% of global GHG emissions¹⁶. Increasing nitrogen-use efficiency is the most beneficial strategy to reduce emissions, and polymer coatings are an efficient means to control the release of nutrients in fertilizer formulations so that they better coincide with the plant life cycle¹⁷. For example, it is estimated by the Japanese Association of Agriculture and Forestry Statistics that between 1976 and 2018, 2.3 million tons of polyurethane- and PE-coated fertilizers were used in Japan, and these helped to control nutrient release, increase nutrient use efficiency, and reduce nutrient loss¹⁸. The European Chemicals Agency (ECHA) has proposed to ban non-biodegradable polymer coatings for controlled-release fertilizers, and the draft regulation with a transition period of 5 years was discussed by the European Commission in September 2022. Recent developments have focused on biodegradable, non-toxic biopolymers to further optimize efficiency and ensure targeted release. For example, using this strategy, 25% of the recommended nutrient loading has been shown to yield an equivalent photosynthetic performance in

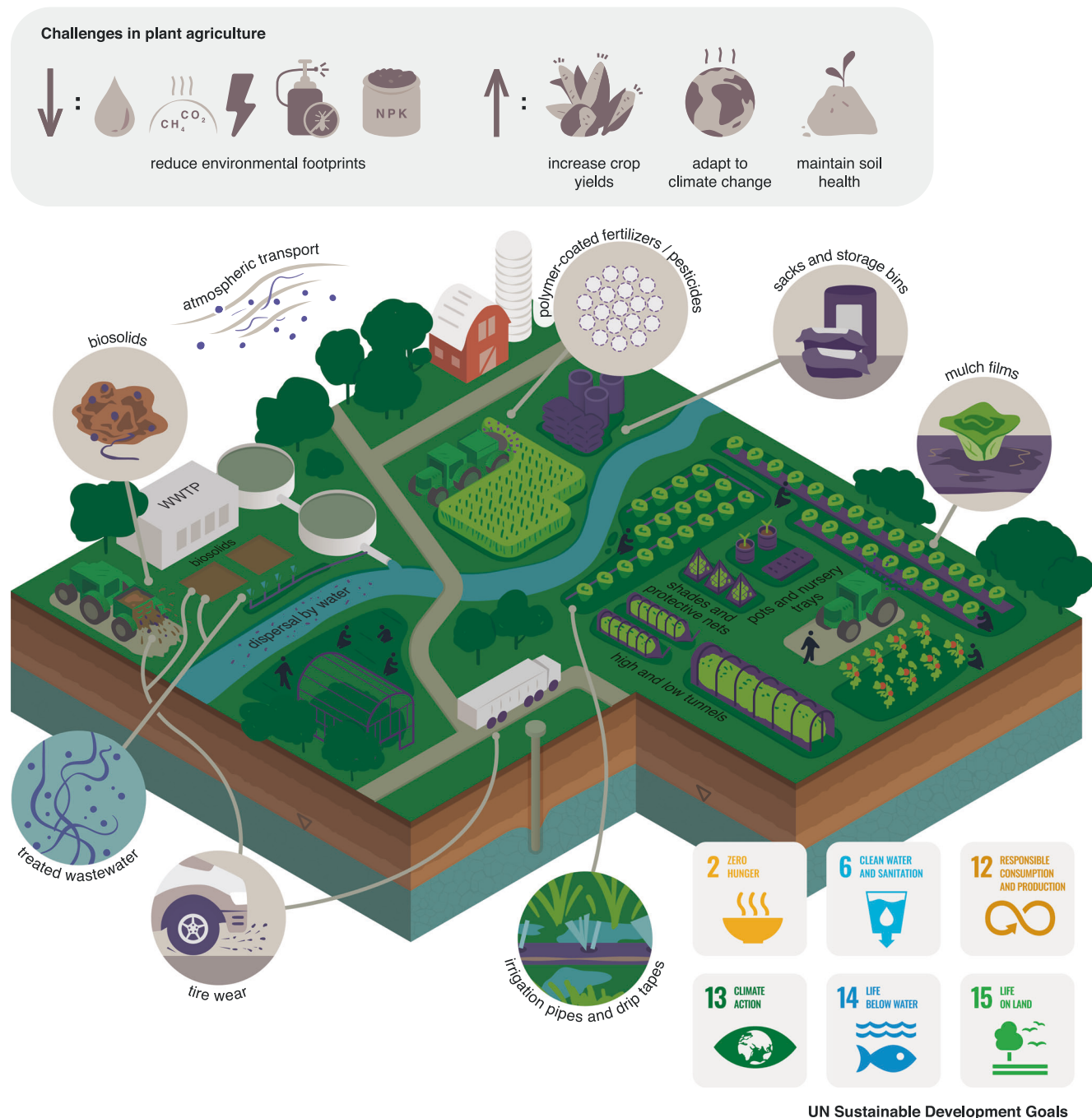


Fig. 1 Applications of plastics in plant agriculture. There are multiple applications of plastics in plant agriculture. Plastics help to reduce irrigation water, pesticide, and fertilizer demand, leading to reduced greenhouse gas emissions and increased crop yield, which impacts several UN Sustainable Development Goals. There are also diverse sources of incidental plastic pollution on croplands; for example, during the dispersal of wastewater biosolids or treated wastewater from wastewater treatment plants (WWTP), including tire wear particles contained therein.

soybean and wheat as did a 100% application rate of the conventional formulation¹⁹.

Water scarcity is a major factor limiting crop growth and yield²⁰. In addition to mulch films, irrigation pipes and drip tapes help to direct precise amounts of water to plant roots, and, hence, to improve water use efficiency. Irrigation equipment is usually made of conventional polymers like high-density PE or PVC. Seedling plugs and nursery pot trays are widely used in crop production. Seedling plug trays allow for efficient germination and optimize plant growth. In addition, some crops require support during cultivation (climbing species, vines). Finally,

nursery pot trays are used for transport, and plastics are used to produce agricultural packaging materials such as fertilizer and storage bags, flexible bulk containers, crates, and containers for pesticides. While the focus of this review is on field applications, a variety of plastics are also used in hydroponic systems and indoor vertical farming for tubing, clamps, grow trays, grids, packaging, and net pots.

Adverse effects and risks

Plastics are an additional stressor that potentially harm essential soil functions. The widespread use of plastics in crop production

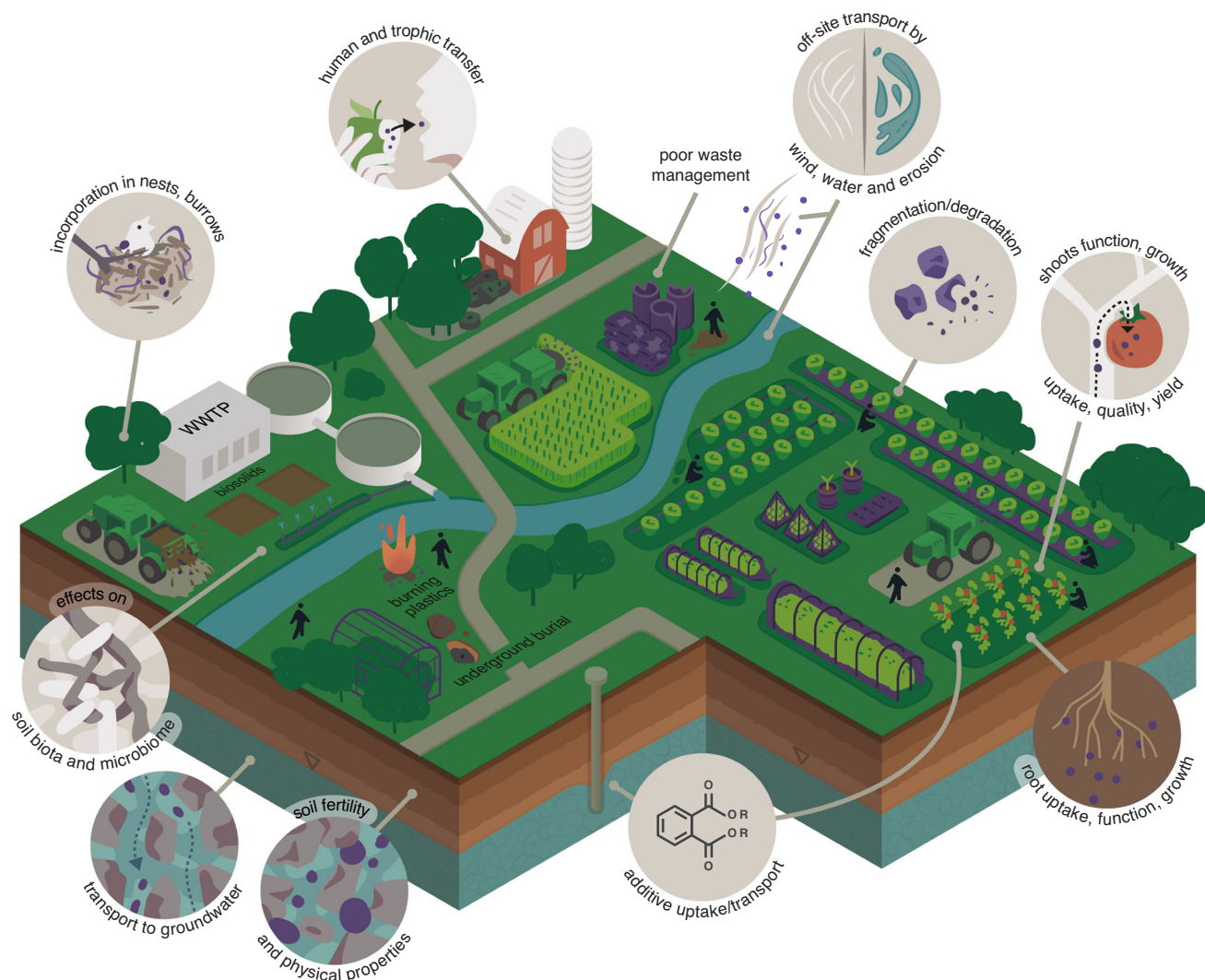


Fig. 2 Adverse effects of plastics in agriculture. The widespread use of conventional plastics in agriculture has led to their accumulation in soils with diverse and long-term effects on crop production, for example, reduced crop quality, and adversely impacted soil health and ecosystem functioning, with unclear uptake potential through the human food chain and impact on human health.

has been reported to affect the physical, chemical, and biological properties of soil (Fig. 2). Due to the persistence of conventional plastics in the environment, plastic fragments will inevitably accumulate in soils over time, disintegrate into MNP and release additives, which may also negatively impact soil health. The effects of plastics on soil properties and fertility are strongly influenced by the properties of the material (i.e., size, morphology, and chemical composition)²¹. The residues of conventional mulch films in the soil can hinder water infiltration, decrease water holding capacity, impact microbial communities and macrofauna, and decrease soil fertility^{22,23}. As a result, plant growth and yields may be negatively impacted. Negative impacts have been observed at high plastic concentrations (>240 kg/ha)²⁴, where conventional, non-biodegradable plastic mulch films are repeatedly incompletely removed or tilled into the soils.

Plastics are not the only stressors acting on agricultural ecosystems. Many global change stressors are acting concurrently on agricultural ecosystems²⁵, including physical (e.g., warming), chemical (e.g., pesticides), and biological (e.g., invasive plant species or weeds) stressors. Recent work suggests that the combined pressures and the high number of factors acting on agricultural soils can lead to unpredictable effects in the soil ecosystem^{26,27}. The comprehensive effects on soil systems are

driven by multiple natural and anthropogenic factors and dissecting the important interactions with plastics should be a major research effort²⁸.

Soil particle transport and the co-transport of contaminants have been intensively studied and the governing processes are generally transferable to the transport of plastics in soils and to aquifers²⁹. The properties of the plastics themselves (e.g., size, morphology, surface properties), the properties of the soils (physicochemical and hydraulic conditions, biogenic activity), and the soil-plastic-particle interactions determine their transport in soils and groundwater³⁰. However, recent studies show that, in general, plastics are not a dominant vector for contaminant co-transport to deeper soil layers and groundwater³¹. Soil erosion is anticipated to be an important diffuse pathway for plastics to water bodies³² and plastic fragment size is a sensitive parameter in determining erosion³². Atmospheric plastic transport involves (re)suspension–deposition cycles³³ and due to the lower density of plastics compared to soil particles, plastic concentrations in wind-eroded soil materials might be higher than in the original agricultural soils³⁴.

Plastics are taken up by biota and can disrupt microbiome functions. Plastic particles of sufficiently small size can be taken

up into plants, while larger plastic fragments can attach to root surfaces and be consumed by humans in the case of root crops. Crossing plant barriers and translocation through plant cells is limited to nanoplastics ($<1\ \mu\text{m}$), with the apoplastic pathway being the main transport pathway, although the symplastic route cannot be excluded based on available data³⁵. Nevertheless, most plant uptake studies so far have been performed under hydroponic conditions and not in soil systems³⁶. While the trophic transfer of plastics in terrestrial food chains has been demonstrated, there is no clear information on the magnitude of the direct transfer of plastics from crops to humans through food.

A better understanding of the potential of plastics and leached plastic additives to accumulate in plants and enter the food chain is critical for food safety. The effects of plastics in food on human health are unclear and further research is needed on their translocation across biological barriers and uptake into organs, as well as the adverse health effects that may result^{37,38}.

Plastics have been shown to negatively affect the growth of crops and animals (e.g., ciliates, flagellates), and cause soil bacterial community structure dysbiosis. Physiological and biochemical impacts on seeds, shoots, and roots of crop plants (e.g., cucumber, wheat, rice, beans) following exposure and uptake have been demonstrated in hydroponic systems at higher concentrations^{39,40}. The effects of plastics on soil microfauna (e.g., *Caenorhabditis elegans*), mesofauna (e.g., *Folsomia candida*), and macrofauna (e.g., snails, earthworms) include adverse behavioral (mobility, avoidance), physiological (growth, reproduction, mortality) and biochemical impacts (oxidative stress, energy metabolism)²⁹. Plastic particles induce microbiome compositional and functional disruption⁴¹. The uptake and effects of MNP also vary strongly, and distinguishing individual and mixture effects is challenging⁴². Nonetheless, one caveat with respect to available research results is that experiments often use hydroponic systems and high concentrations of 0.1–10% w/w, which are not representative for soils or field conditions⁴³.

Leaching of additives from plastics increases the chemical burden on soils. Aging and fragmentation of macroplastics into MNPs is expected to not only contribute to the dispersion of plastics in the environment but also enhance the release of additives and their degradation products. Leached additives can be transferred to biota and plants⁴⁴, or sorb to soil. Time scales over which chemicals are released vary widely. For example, the release of hydrophobic phthalates such as di(2-diethylhexyl) phthalate (DEHP) can extend over centuries due to slow diffusional aqueous boundary mass transfer, while more polar phthalates such as di-butyl phthalate are released within days^{45,46}. The consequences of a long-term release of chemicals due to the degradation of plastics in soils are unknown⁴⁷ and the plastic toxicity debt needs to be reduced. Additives such as phthalates and bisphenols are known to have direct toxic effects from endocrine-disrupting properties⁴⁸, while others such as the transformation products of the tire antioxidant 6PPD have shown toxicity across multiple environmental compartments⁴⁹. Following their release, phthalate esters and additives from tires may accumulate and be metabolized in edible plants, serving as a point of entry to the human food web. The application of agrochemical formulations stored in fluorinated polyethylene containers used in agriculture and fragments from discarded plastic containers in agricultural soils may be a source of contamination of per- and polyfluoroalkyl substances for agricultural soils⁵⁰.

Sustainable use of plastics in plant agriculture

The sustainable use of plastics in plant agriculture will require alignment with the “3 R” waste hierarchy concept of reducing,

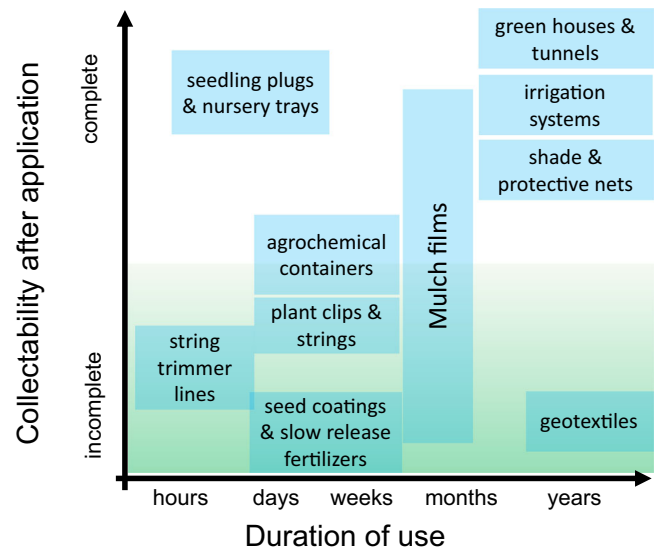


Fig. 3 Collectability and duration of use of plastics in agriculture.

Collectability after use and duration of use. For plastics that can be collected in their entirety (i.e., complete collectability after use), priority should be given to reuse and recycling. Innovation in the material design must be geared towards complete collectability, recyclability, and reuse. Plastics that cannot be collected in their entirety after use should be fully biodegradable, e.g., seed coatings, slow-release fertilizers, and thin ($<20\text{--}25\ \mu\text{m}$) mulch films. Material design must be geared towards ensuring sufficient stability of the plastics during use, while allowing for biodegradability after use.

reusing, and recycling plastics as preferred options over disposal after use. Greater scrutiny is required for plastic applications that only have non-circular end-of-life treatment options. Adding to these principles, two additional criteria can provide guidance in determining sustainable use strategies and identify after-use treatment options for plastics in agriculture: (i) collectability of plastics after use and (ii) duration of plastic use. For applications that allow complete collection after use, reuse, and recycling are the preferred treatment options, irrespective of the usage duration (Fig. 3). For these applications, advances toward sustainability must be made by using plastics that do not chemically weather and fragment, and by implementing technologies that ensure complete collection after use, especially for plastics used below ground level, such as irrigation pipes.

For applications where plastics cannot be completely collected after use, or where the collected plastic is too degraded or soiled to be reused or recycled, less toxic and biodegradable polymers should replace conventional persistent polymers, especially in cases with a short duration of use such as mulch films or non-biodegradable polymer coatings for controlled-release fertilizers. The biodegradable plastics must be able to function throughout the entire period of application and, at the same time, be completely converted to CO_2 and microbial biomass in the soil within a specified time period, at the end of their life. These applications include, but are not limited to seed coatings, slow-release fertilizers, thin mulch films, as well as geotextiles (Fig. 3).

Collection after use, reuse, and recycling should be prioritized.

The rates of reuse and recycling for agricultural plastic waste are currently very low ($<10\%$) and need to be substantially increased⁵¹. In cases where the reuse of agricultural plastics is challenging, it is imperative to prioritize the elimination of problematic types. For instance, we strongly advocate for the prohibition of oxo-degradable mulches, the restriction of harmful

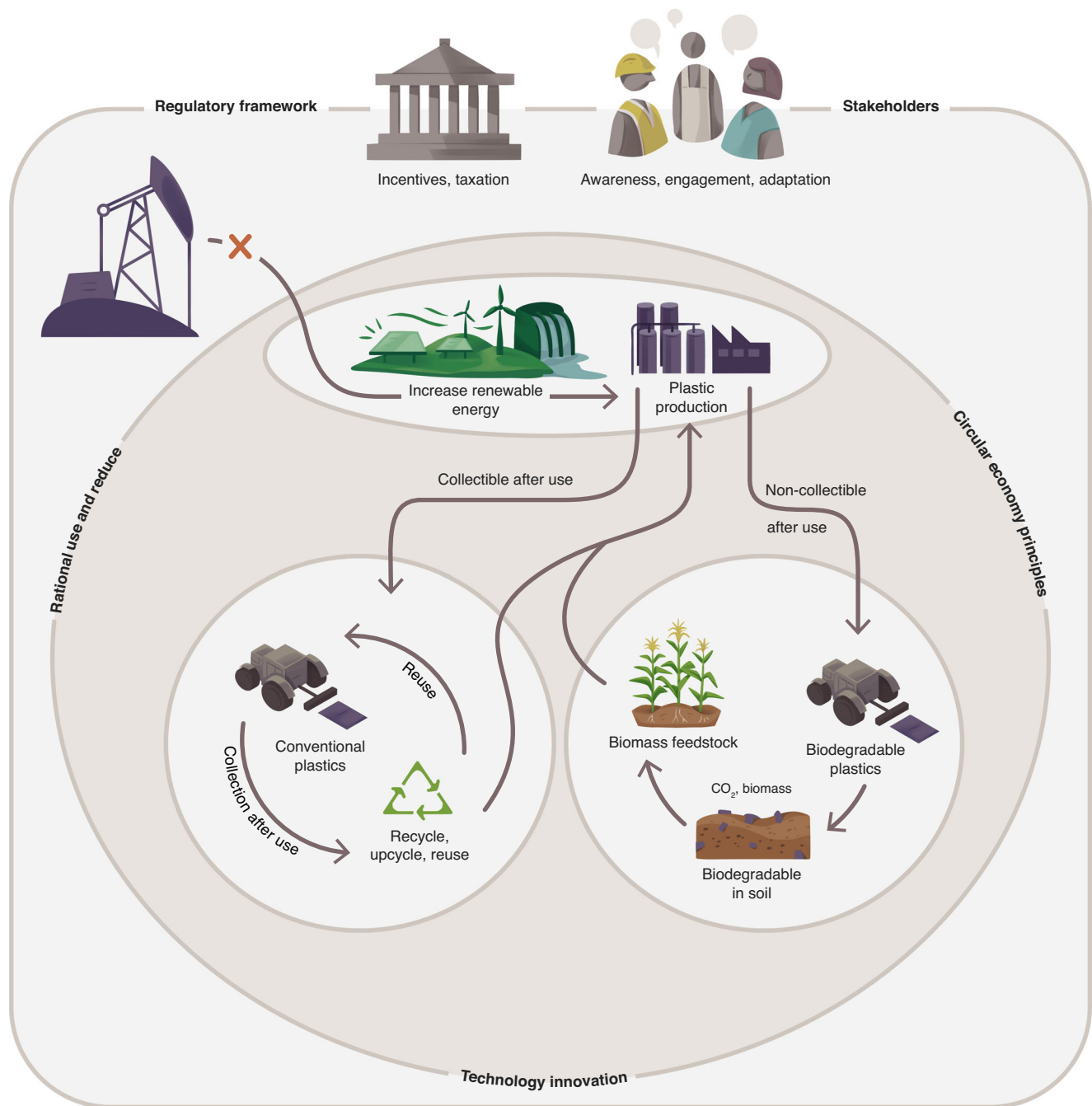


Fig. 4 How to use plastics more sustainably in plant agriculture. The regulatory framework and targets for the use of plastics in plant agriculture should be strict and legally binding for all UNEA member states. Awareness, engagement, and adaptation are needed to reduce the use of plastics, involving all stakeholders. A shift to renewable energy sources is needed in plastics production to reduce greenhouse gas emissions. Plastics that can be collected in their entirety after use must be recycled, upcycled, or reused. Rates of reuse and recycling need to be substantially increased. Non-collectable plastics must be fully biodegradable in the soil. To ensure sustainable use, technological innovation, circular economy principles, rational use, and reduced use of plastics are required in all areas of agricultural food production.

polyvinyl chloride usage, and the phasing out of intentionally added microplastic applications (see Fig. 4). Therefore, plastics that are not substantially damaged during routine use should be prioritized for reuse on farms, or through collection, facilitated by programs such as extended user responsibility and reverse logistics^{52,53}.

After collection, non-reusable agricultural plastics should undergo a rigorous cleaning process to eliminate soil contamination and other debris before being sent to materials resource facilities (MRF) for recycling. Some agricultural plastics, such as

agrichemical containers, can be effectively decontaminated and reused⁵⁴. Economic feasibility and environmental benefits can be realized by collecting bale wrap films, which are typically less soiled compared to ground-collected mulch films⁵⁵. Additionally, plastic mulch films can be recycled through pyrolysis to produce fuels or polymers, as the presence of soil or other contaminants does not affect this thermal treatment process⁵⁶. To facilitate agricultural plastic recycling, it is essential to establish dedicated programs either through legislation or voluntary initiatives, encompassing well-coordinated and widely accessible collection

facilities. If agricultural plastic waste is excessively contaminated or physically degraded, it should be appropriately disposed of in a landfill, but this option is the least desirable.

MRF sort and process plastics using mechanical means such as shredding followed by melting and re-extrusion to generate downcycled plastic resins or asphalt. Chemical processing techniques such as pyrolysis, dissolution, re-extrusion, chemical or enzymatic methanolysis, or gasification can produce a range of up-cyclable products. These include feedstock chemicals such as aromatic chemicals and olefins, high-quality plastic resins and original plastic monomers, methanol, syngas, and transportation fuels⁵⁷. There is substantial research on different upcycling technologies for waste plastics such as hydrogenolysis of plastics to yield lubricants and oils, and enzymatic treatment or chemical and enzyme-assisted mechanochemical processing of PET to yield the primary monomers^{58,59}. Newly developed zirconium-catalyzed C-H and C-C aluminolysis processes might make it easier to recycle and upcycle plastics, or make them biodegradable⁶⁰. Technologies are mostly in operation commercially, with some at industrial demonstration stages, but they should be able to address recycling of the various types of plastics used in agriculture. Although processing techniques are specific to plastics and their mixtures, plastics used in agriculture are in general recyclable. In fact, plastic film waste from greenhouses has been directly used for laboratory demonstration of catalytic cracking to enable feedstock recycling⁶¹. Furthermore, a closed-loop supply chain analysis for bale wrap collection, sorting, and recovery in Finland showed substantial economic savings and reduced global warming potential⁵⁵. A closed-loop recycling philosophy could also be employed for self-immolative polymers. Although the feasibility of a larger-scale application of breaking down and then re-constituting the polymers for many cycles has not yet been fully explored, recent reviews have highlighted the interest and importance of such materials^{62,63}.

The types of chemical additives present in plastic products can substantially affect recyclability. For mechanical recycling, the presence of certain additives and their transformation products affects the safety and marketability of the secondary plastics. For instance, when plastics with different pigments are mixed, this may result in the coloration of the secondary product, restricting its further utilization⁶⁴. Toxic by-products may also be produced if recycling is performed at unsuitable temperatures, which may lead to an accumulation of these substances and a reduction of the suitability of the plastics in a circular economy⁶⁵. Therefore, it will be essential to integrate management of chemical additives into the global Plastics Treaty in order to reduce their complexity in plastic mixtures⁶⁶. Technologies such as dissolution-based recycling can be designed to selectively remove problematic additives and components, but these techniques are likely to be more expensive and energy intensive⁵⁷.

Although plastic wastes in agriculture have the advantage of providing a relatively homogeneous stream of compositionally viable materials for recycling, within concentrated geographic locations, there are nonetheless technical obstacles. Thin films that are too brittle or weathered cannot be collected in their entirety after use. Exposure to heat, solar UV radiation, soil, and agricultural chemicals can degrade some of the plastics and make recovery difficult or only allow down-cycling by mechanical processing^{67,68}. Furthermore, the seasonal variability linked to the generation of agricultural plastic wastes makes it challenging for recycling facilities to create stable secondary markets.

For plastic mulch films, the major obstacle to recycling is the contamination with soil adhering to the plastics when they are recovered from the field after harvest. Soil and plant residue contamination can account for up to 80% of the total weight of plastic materials, while existing recycling facilities require

contamination levels below 5%⁵⁶. Therefore, it is crucial to develop effective techniques for removing soil and plant residues from plastic films. These techniques may involve mechanical or manual cleaning processes and minimizing soil adhesion. Ideally, they should be implemented directly on-site at the farm to reduce transportation costs, using mechanical rotary brooms or manual methods on dry days, which facilitate collection and recyclability⁵⁶. Consequently, there is a pressing need for improved techniques and standardized methods for post-use collection, decontamination, and handling to enhance recycling^{67,69,70}.

Plastics that cannot be collected in their entirety after use should be biodegradable. Biodegradation is considered a viable end-of-life option for agricultural plastic applications in which the plastics cannot be collected from the field in their entirety after use, as well as for cases where the collected plastics fractions are too weathered or soiled to allow for reuse and recycling (Fig. 4). Applications where collecting plastics after use is challenging, if not impossible, include for example thin agricultural mulch films, abraded plastic from string trimmers and plant clips, and geotextiles^{71,72}. In these cases, complete biodegradation of the plastics to CO₂ and biomass ensures that no residues will accumulate in soils. The use of polymers with cleavable bonds such as esters or amides⁷³ can allow biodegradation to be tuned to occur at the end of life. Modifications of the backbone chemistry of existing polymers to render them biodegradable is an emerging field of research. For example, a novel bio-based polyester-2,18 exhibits high-density PE-like properties, but is readily hydrolyzable by naturally occurring enzymes⁷⁴. Conversely, using prooxidant (i.e., “oxo”) additives that trigger polyolefin breakdown upon thermal or photochemical activation is an inadequate technology that has failed to ensure plastic biodegradability⁷⁵.

It will be necessary to carefully scrutinize elevated inputs of biodegradable plastics in order to ensure that they indeed biodegrade in situ over a reasonable, defined timeframe. Presently, the biodegradability of plastics in soil is assessed in laboratory soil incubation studies under oxic conditions that couple respirometric analyses of polymer conversion to CO₂ under constant humidity and elevated, constant temperatures (typically at 20–25 °C). These conditions are expected to favor biodegradation compared to the field, and research is needed to determine the extent to which biodegradation data obtained in the laboratory are transferable to more realistic field conditions. Furthermore, greater efforts need to be directed toward monitoring the concentrations of biodegradable plastics in situ over long periods (~years) and following repeated applications, in order to determine steady-state concentrations in soils and to identify the key factors affecting biodegradation rates. Analytical techniques for accurately quantifying low concentrations of polymers in soils are sorely required⁷⁶. Stringent biodegradability standards that define testing conditions and stipulate the required mineralization timeframe for the incubation are key to ensuring biodegradability. Such a standard exists for biodegradable mulch films (EN 17033:2019), which stipulates 90% of polymer mineralization in soils over a two-year laboratory soil incubation at 20–25 °C, as assessed by established methods (ASTM-D5988). Scientific advice will be critical to future regulatory efforts for both the plastics and their additives. Indeed, additives are expected to be readily released during biodegradation, resulting in potentially higher concentrations in soil porewaters compared to additives leached from conventional plastics, highlighting the need for using environmentally benign additives only.

Presently, biodegradable plastics are 20–80% more expensive to produce. Although high costs could considerably reduce their

acceptability⁷⁷, the costs of collection, recycling, or disposal needed for conventional plastics are generally not included in such calculations. Furthermore, negative impacts on soil productivity, resulting from the long-term accumulation of conventional non-biodegradable plastics in agricultural soils have not been valorized.

Mandatory use of environmentally benign additives is necessary to reduce toxicity burdens. Plastic (bio)degradation and the concomitant leaching of additives into soils are of concern. Plasticizers have been intensely scrutinized due to their high loading in many plastic materials (e.g., typically about 30–35 wt% for PVC). One of the most common plasticizers used, DEHP and its monoester metabolite, were shown to act as endocrine disruptors⁷⁸, leading to its ban in children's toys⁷⁹. Understanding degradation pathways for phthalates has led to the development of alternative plasticizers, such as the non-phthalate di(2-ethylhexyl) terephthalate and others such as di(isononyl) cyclohexane-1,2-dicarboxylate (DINCH). These alternative plasticizers have been grouped under the umbrella of “green plasticizers”, and, along with their metabolites, are viewed as being non-toxic. However, it has recently been discussed that DINCH might have a metabolic toxicity to aquatic organisms⁸⁰, and the challenges of assessing whether a plasticizer is truly “green” has been highlighted in several reviews^{81–83}. While some plasticizers may use renewable feedstocks, their syntheses often involve hazardous chemicals, underscoring the importance of performing life cycle analysis to avoid regrettable substitutions⁸⁴.

More than 10,000 chemicals are used in the production of plastics, many of which are substances of concern based on persistence, bioaccumulation, and toxicity criteria⁸⁵. A transition towards sustainable plastic applications will require transparency and accessibility of information regarding the chemicals used in the production of plastics. For some chemicals such as phthalates, there is extensive literature on their release from plastics and their uptake by soil organisms. However, for most plastic products used in plant agriculture this information has not been thoroughly documented and fate studies are needed^{44,86}. To address plastic pollution concerns holistically, including the diversity and complexity of chemicals added to plastics, chemicals should be included in UNEA-5.2, which will foster innovation in the design of agricultural plastics⁶⁶.

Sustainability innovation should be linked to social processes. Public concern about plastic waste in general, and microplastics specifically, is high^{87,88} and shifting from marine environments to soils, air, and human health^{21,87}, even though evidence for the adverse effects of MNP on human health is currently lacking⁸⁹. Food safety is a particularly sensitive topic. Microplastics in food were among the top ten food safety concerns in Europe in 2022⁹⁰. Such public concern can affect consumer demand and the reputation of producers.

There is a gap in research on risk and benefit perceptions of plastic in the context of food from the perspectives of stakeholders in the entire supply chain, including producers, processors, packaging industry, and end users, and a gap in research on relevant education and training programs to change current practices. Relevant factors may include convenience, cost, industry communications, and regulatory frameworks, but also awareness of the systemic and long-term risks of plastics to ecosystems and future livelihoods, all of which could predict stakeholder willingness to make changes.

For farmers specifically, recent work showed that 80% of Irish farmers reported an increase in plastics use and 88% reported concern about negative environmental impacts⁷⁰. Disposal

practices varied widely, depending on materials in use, reported knowledge, perceptions of costs and facilities, and education level, but links between general education level and behavior are inconsistent⁹¹.

Moreover, while some attempts are being made to provide specific training, for example, “Plastics in Agriculture Lessons—Preventing Plastic Pollution”⁹², to the best of our knowledge, there are currently no comprehensive programs regarding the life cycle implications of plastics and practical know-how on new skills and practices. Such new training programs should be addressed to diverse stakeholders and cover all life cycle phases of plastics. Crucially, such programs should not just teach factual knowledge, but integrate best practice in line with psychological approaches that emphasize the importance of factors beyond mere knowledge, such as motivations, norms, and values in predicting behavior change⁹³.

Thus, the reduction of plastic pollution is a question of perceptions and behavior, as well as of the materials available, production practices on farms, and contextual constraints. It would be useful to quantify the variance attributable to different behavioral practices to determine the importance of this particular lever in agriculture and along the food supply chain. Future work to reduce plastic pollution should build on existing approaches from soil conservation behaviors and environmental stewardship, and must better capture the interplay between environmental and social processes^{94,95}.

Renewable energy production and sources. The use of plastics involves larger-scale aspects related to the sustainability of the production process. For example, while the production and use of plastics account for 4.5% of global greenhouse gas emissions, 96% of these emissions are attributable to their production using non-renewable resources. These emissions can be allocated to fossil fuel extraction (10%), electricity generation (32%), polymer production and manufacturing (41%), and other sectors (17%), underscoring the importance of reducing the production of primary plastics⁹⁶. In cases where the use of plastics in agriculture is essential, effective measures to reduce the carbon footprint of the plastics include transitioning to renewable energy, increasing energy efficiency in the fabrication process, and promoting the use of bio-based plastics. The use of agricultural plastics from non-renewable sources should be disincentivized (e.g., with environmental taxes), and innovations that promote bio-based materials should be prioritized (e.g., with tax subsidies). Although the raw materials for bio-based polymers can in some cases be produced from waste materials such as crab shells, shellfish waste, or insect exoskeletons for chitosan⁹⁷, a substantial increase in demand for the biological raw materials may result in undesired competition with food production^{98,99}. Increased demand could also substantially intensify pressure on marine and terrestrial ecosystems. Therefore, the substitution of fossil-based materials with bio-based materials must be carefully evaluated, by considering the entire life cycle, to avoid unwanted concomitant risks.

Regulatory frameworks and incentives. There is consensus among scientists, members of government, non-governmental organizations, and stakeholders that sources of plastic pollution must be addressed across many areas of society, including plant agriculture. Achieving a more sustainable use of plastics in plant agriculture will require science-based decisions built on circular economy principles, including innovations in material design, collectability after use, reuse, and recycling, and changes in usage practices (Fig. 4).

Currently, plastic products in agriculture cannot be easily eliminated or replaced with non-plastic materials without incurring serious environmental and societal costs¹⁰, and alternatives to plastics with a smaller environmental footprint are not currently widely available. However, commitment to increasingly ‘plastic-free’ agriculture could stimulate technological innovation¹⁰⁰. Agricultural actors can be rewarded with tax breaks or certifications if they adopt more sustainable plastic use practices, including the collection of used plastics for reuse and recycling and the use of fully biodegradable plastics, where appropriate. Such incentives could encourage rapid and widespread adoption. Socio-economic frameworks that disincentivize the use of non-collectible conventional plastics would facilitate such a transition in the agricultural sector. However, as it is imperative to minimize negative economic impacts on farmers, clear policy guidelines are required to ensure a just transition. Global production patterns are diverse, and thus, any approach needs to consider regional and cultural differences in food production. Such a holistic approach must include rational use, optimization of the collection after intended use, reuse, and recycling, innovative design for environmentally benign materials, modification of agricultural practices, and improved management strategies to reduce environmental pollution originating from plastics and associated chemicals. Circular economy principles can provide a framework for specific measures to minimize the environmental footprint and contribute to the sustainable use of plastics in plant agriculture¹⁰¹.

Current negotiations to develop and implement a legally binding global plastics treaty within the United Nations Environment Assembly (UNEA-5.2) aim to curb plastic pollution, including pollution from agricultural plastics^{6,9,102}. Achieving this aim requires an international regulatory framework that considers the full life cycle of agricultural plastics. Targets should be strict and legally binding for all UNEA member states¹⁰³ and regularly adapted to the current state of research. Governments and non-government stakeholders must regularly monitor and report on the use of plastic in agriculture in order to assess the implementation of UNEA-5.2 goals¹⁰⁴ and the performance of actions taken. In addition, an inventory of all plastics used, collected, reused, and recycled would need to be created. Data would be collected by the sector and reported at the national level to establish standardized plastic pollution monitoring databases and track compliance with reduction targets. Monitoring and reporting of plastics used in agriculture with detailed post-use treatment analysis will help ensure that reduction strategies are effective.

Plastics are essential in modern agriculture, aiding in weed and pest control, water conservation, and improving crop quality and yield. While providing agronomic benefits, their production, improper disposal, loss during operations, or abandonment in fields and farmland can pose substantial environmental risks. To mitigate these negative impacts, responsible and sustainable utilization of plastics is imperative. Achieving sustainability in agricultural plastic usage necessitates a comprehensive approach, encompassing rational use, technological advancements in reuse and recycling, adoption of less toxic and biodegradable materials, education and behaviour change, social and economic incentives, and legislative enforcement.

Data availability

Data sharing is not applicable to this article as no datasets were generated or analyzed during the current study.

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Author contributions

T.H., S.G., and N.T. planned and organized the workshop. T.H. led the manuscript writing and editing. The sections of this manuscript are based on the written input from

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Competing interests

The authors declare no competing interests.

Additional information

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